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ME EN 507

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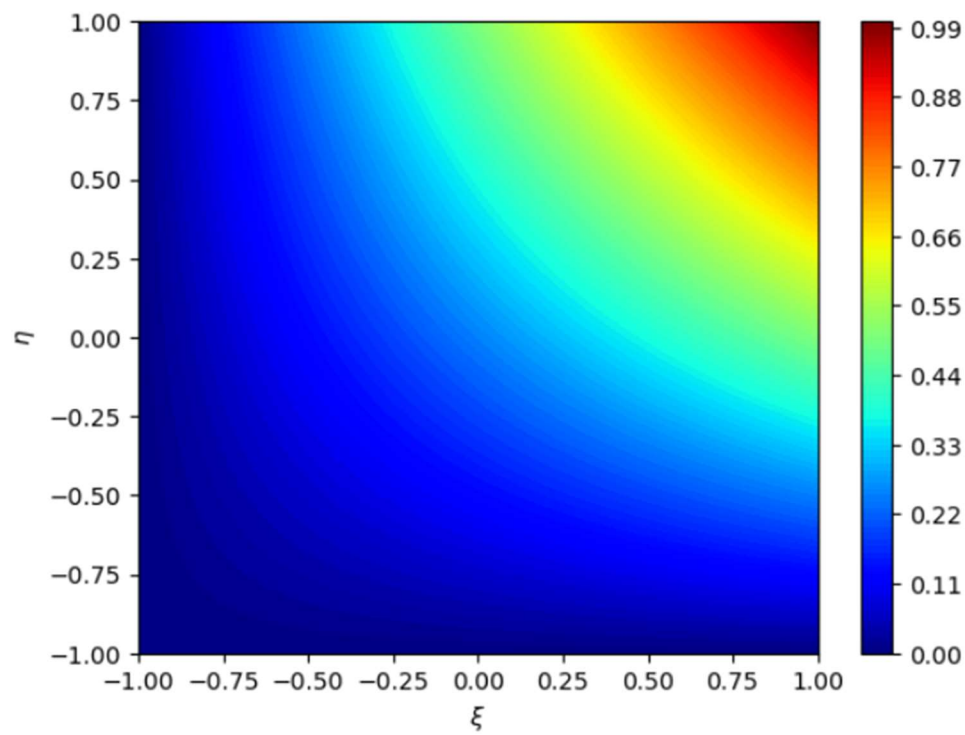
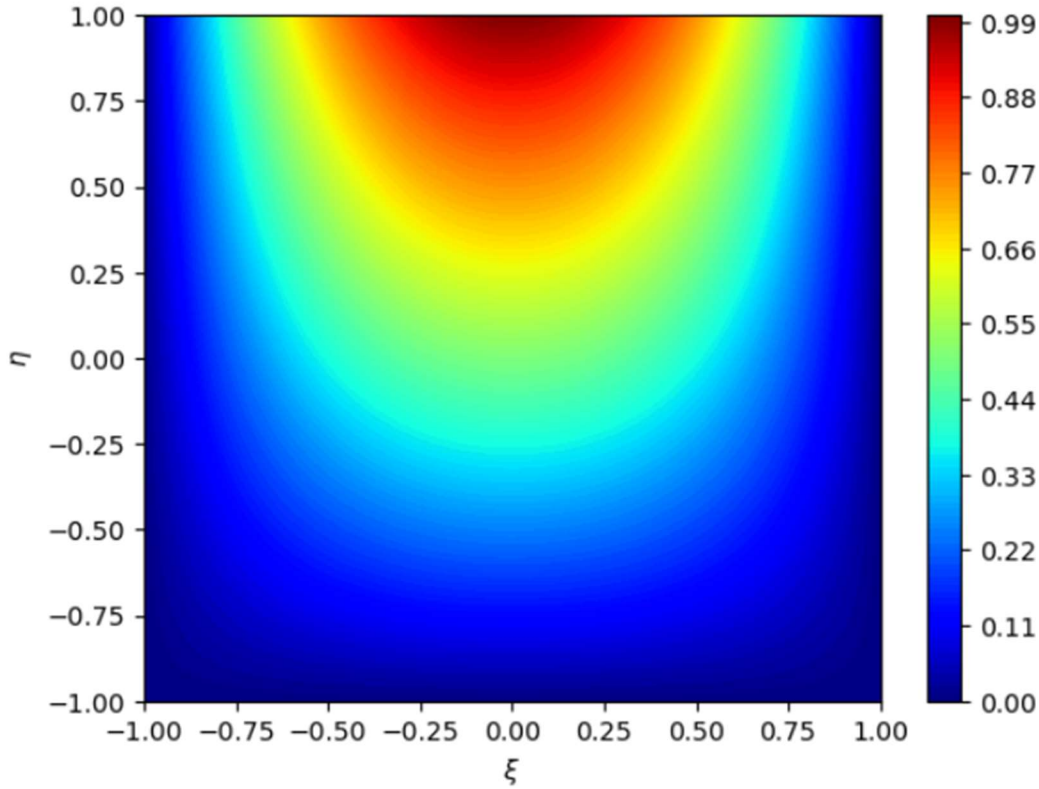
HW 6

```
# higher-dimensional basis function with multi-index
def MultiDimensionalBasisFunctionIdxs(idxs,degs,interp_pts,xis):
    N_a = 1.0 # initialize the product to 1
    for i in range(0, len(idxs)): #Loop over each dimension
        N_a *= LagrangeBasisEvaluation(degs[i], interp_pts[i], xis[i], idxs[i]) #compute the Lgrange b
    return N_a


# higher-dimensional basis function with single index
def MultiDimensionalBasisFunction(A,degs,interp_pts,xis):
    bfs = degs[0] + 1
    horiz = A % bfs
    idxs = [horiz]

    for i in range(1, len(degs)): # iterating
        idxs.append(A // bfs) # equation for converting A into a
        bfs *= (degs[i] + 1)

    return MultiDimensionalBasisFunctionIdxs(idxs, degs, interp_pts, xis)
```

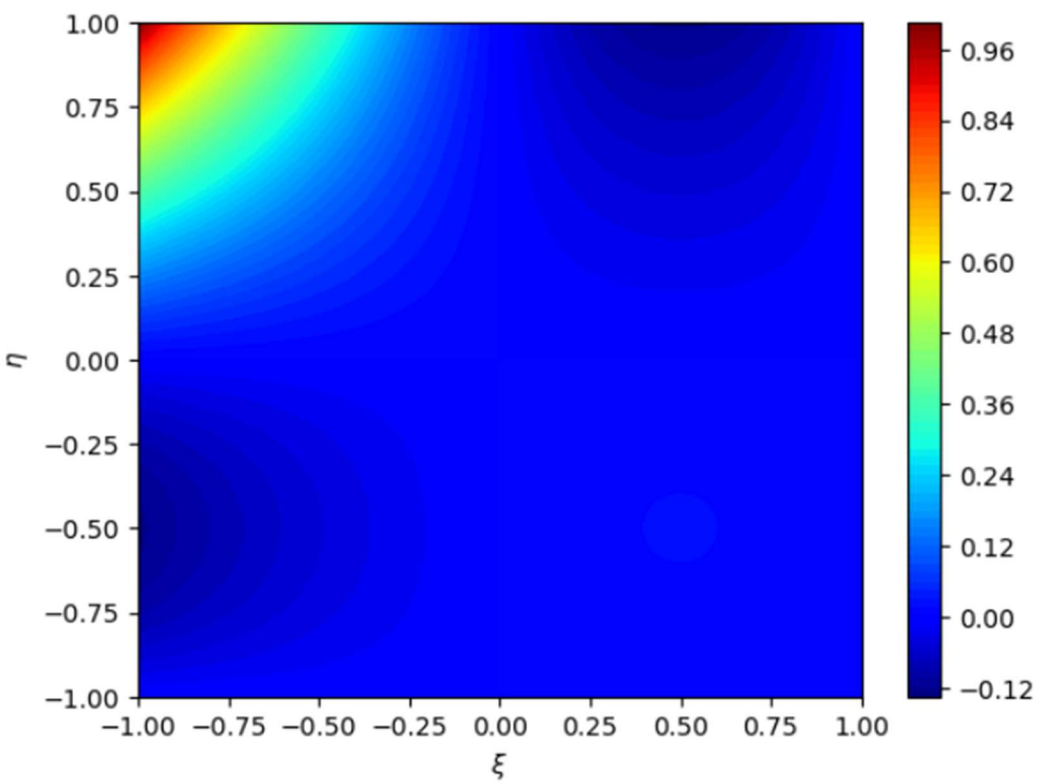
```
# - Polynomials of degree [1, 1]
PlotTwoDimensionalParentBasisFunction(3, [1,1], contours=True)
```



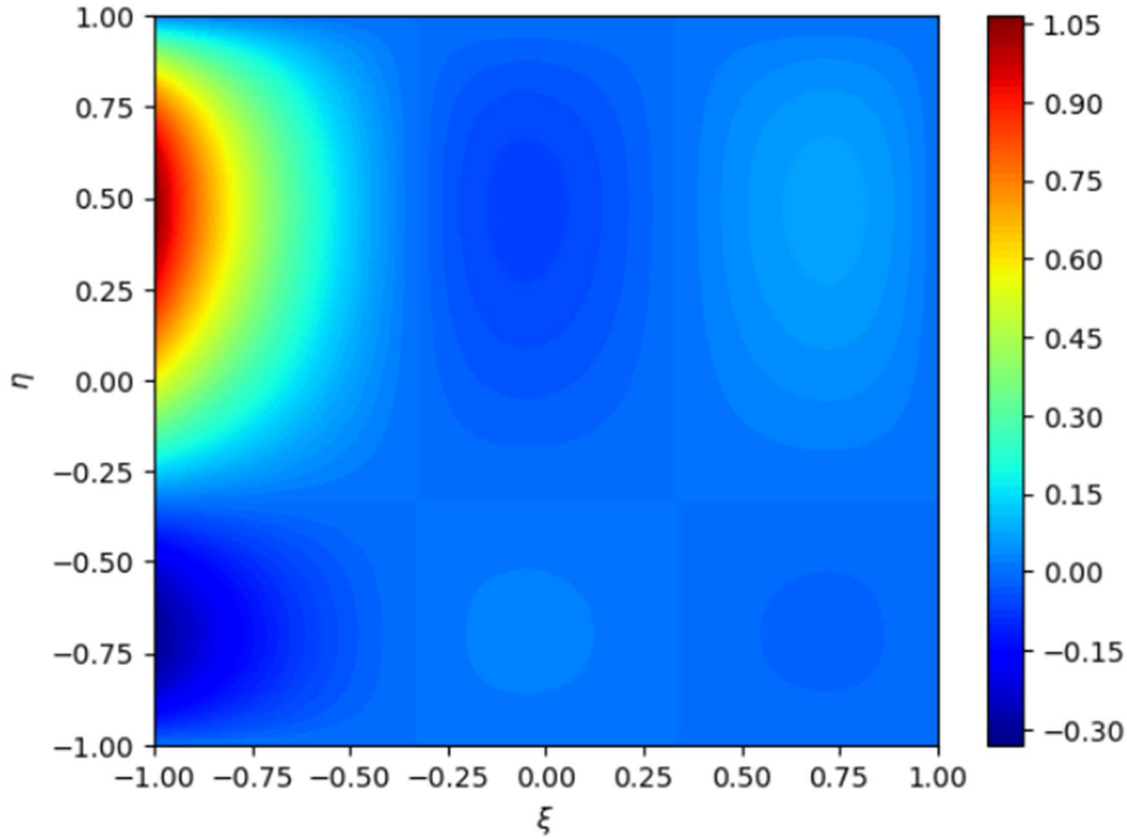
```python
# - Polynomiala of Degree [2, 1]
PlotTwoDimensionalParentBasisFunction(4, [2, 1])
```




```python
# - Polynomials of Degree[2, 2]
PlotTwoDimensionalParentBasisFunction(6, [2, 2])
```



```python
# - Polynomials of Degree [3, 3]
PlotTwoDimensionalParentBasisFunction(8, [3, 3])
```

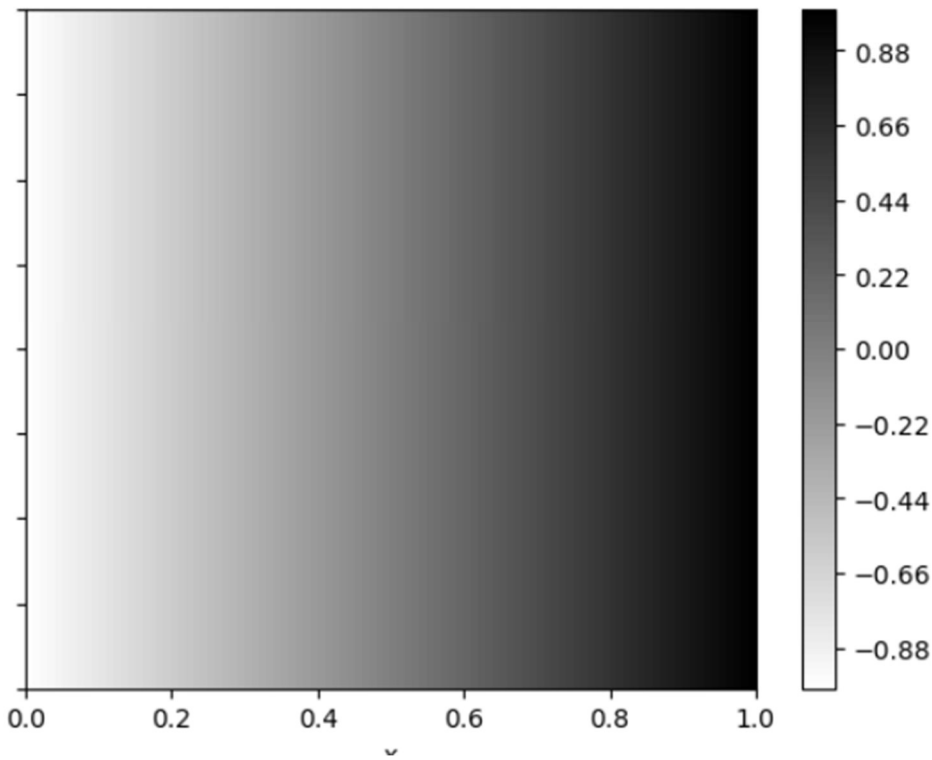


Problem 2

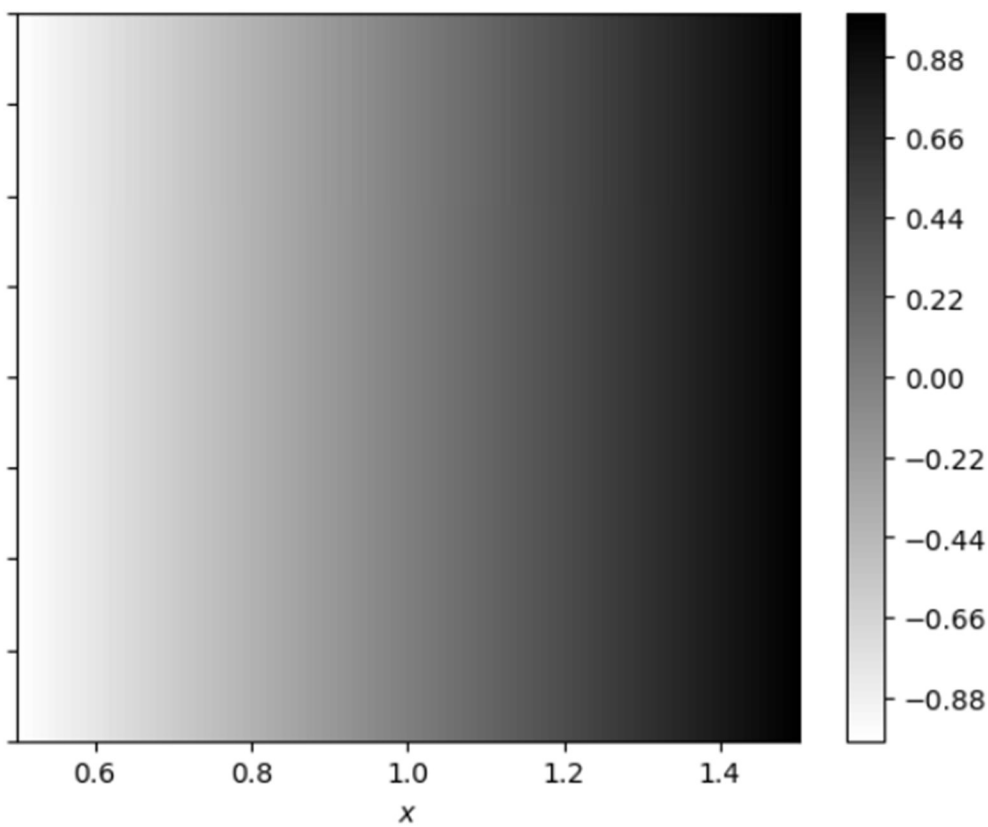
```python
# x(\xi) = \sum_{a=0}^p x_a * N_a(\xi)
def XMap(degs,idxs,interp_pts,xi):
    sum = 0.0 # initialize the sum
    for i in range(0, len(idxs)): #Loop over each dimension
        N_a = LagrangeBasisEvaluation(degs,interp_pts,xi, i)
        sum += idxs[i] * N_a
    return sum
```

The mapping is linear when the degree is one. But is can also be squished or stretched to be linear in any degree. In this assignment I noticed that the mapping was linear in two dimensions if the intervals were evenly spaced.

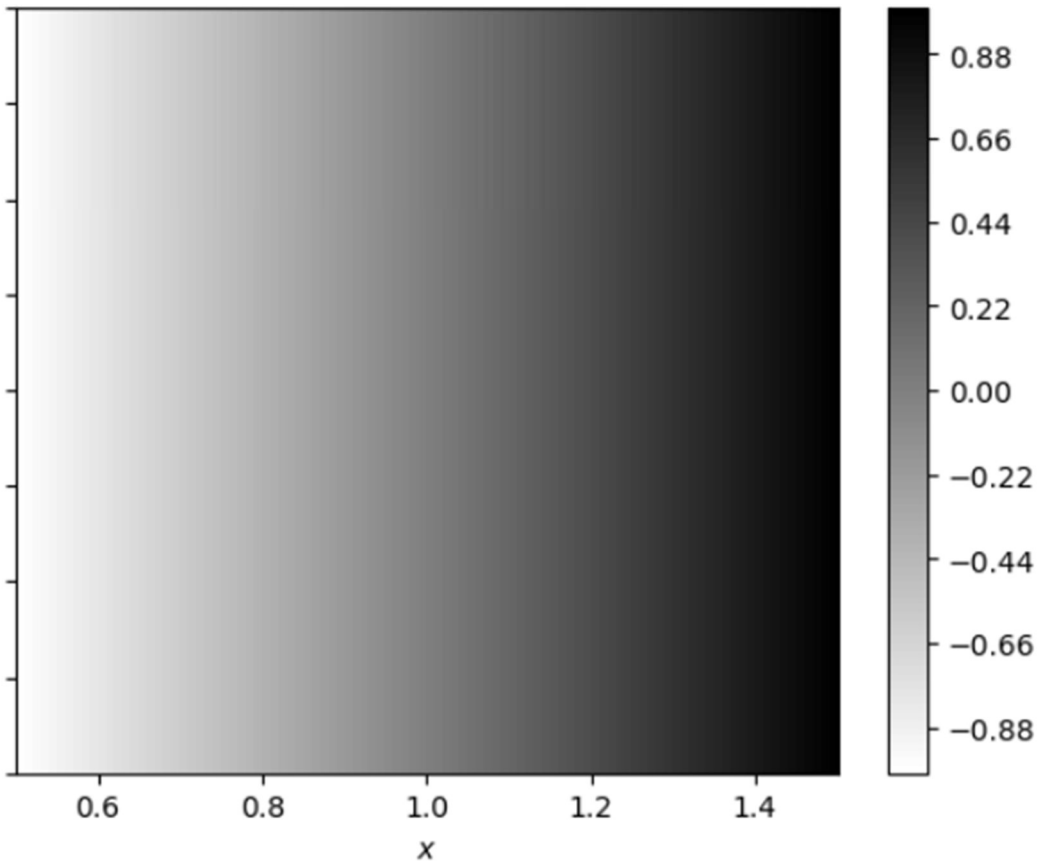
```python
# p = 1, {x_a} = {0, 1}, {xc_a) = {-1, 1}
PlotXMap(1, [0, 1], [-1, 1])
```

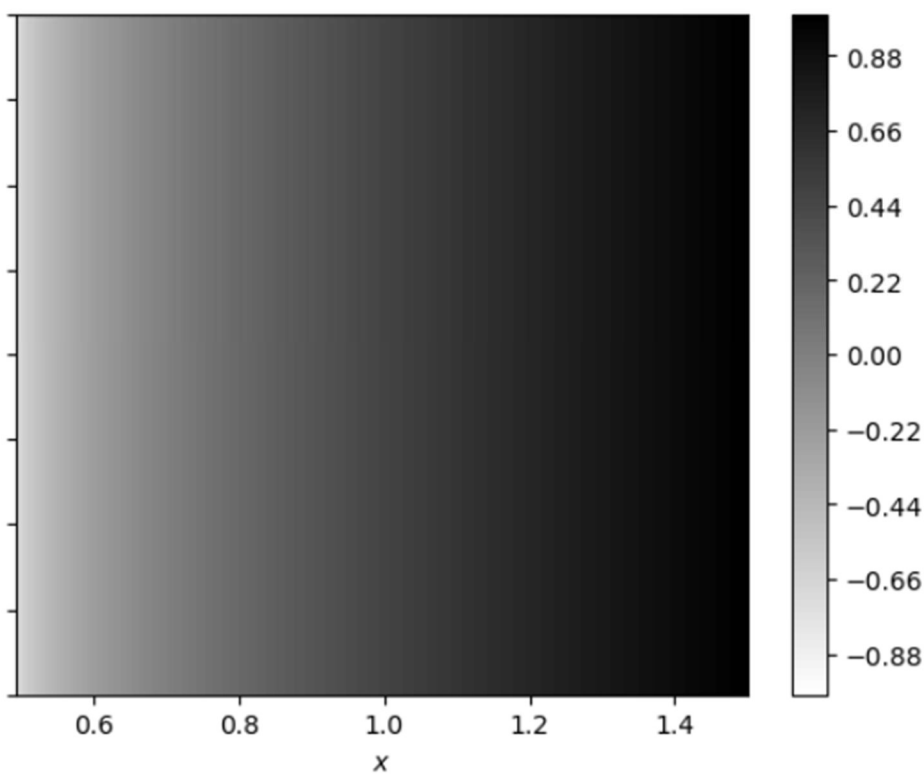



```python
# p = 2, {x_a} = {0.5, 1, 1.5}, {xc_a) = {-1, 0, 1}
PlotXMap(2, [0.5, 1, 1.5], [-1, 0, 1])
```



```
# p = 2, {x_a} = {0.5, 0.7, 1.5}, {xc_a) = {-1, -0.6, 1}
PlotXMap(2, [0.5, 0.7, 1.5], [-1, -0.6, 1])
```




```
# p = 2, {x_a} = {0.5, 0.7, 1.5}, {xc_a) = {-1, 0, 1}
PlotXMap(2, [0.5, 0.7, 1.5], [-1, 0, 1])
```